

3rd year

Signals & Systems Lab

Lab (1)

Plotting signals with Matlab

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Introduction to Signals with MATLAB

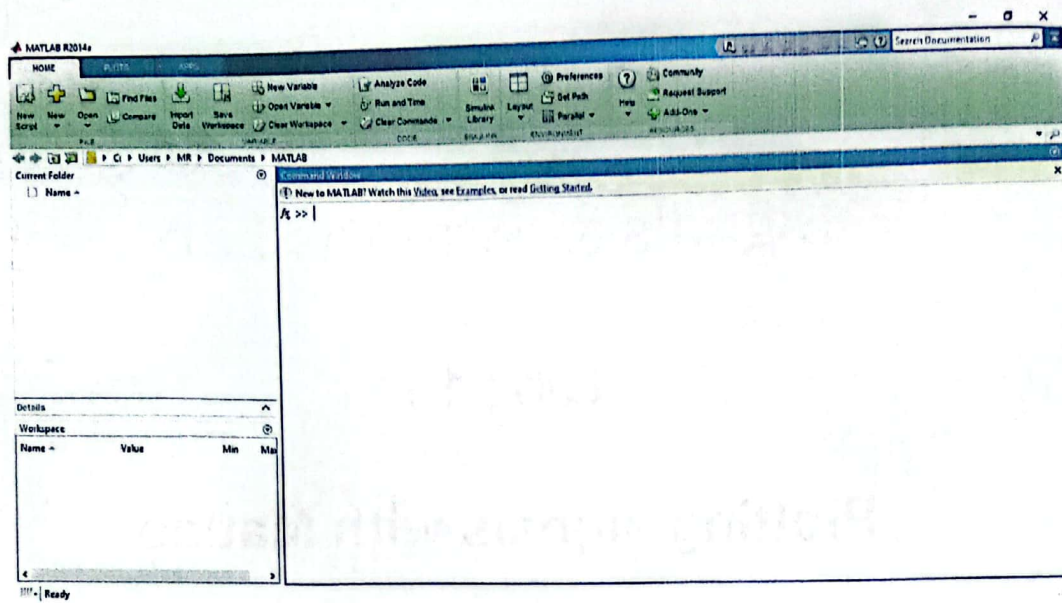
The aim of this lab is to introduce students to the fundamental concept of **signals** and how to represent, analyze, and visualize them using MATLAB. Since signals form the core of Signals and Systems, this introduction provides the foundation for all upcoming experiments. No prior deep knowledge of MATLAB is required; only the basics of operating the environment and executing commands will be used.

There are a few important points for this lab:

1. In MATLAB, a signal is usually represented as a vector (for discrete-time) or a function (for continuous-time). Understanding how to define and manipulate vectors is the first step in signal processing.
2. Visualization is a key part of signal analysis. MATLAB's **plot** functions will be frequently used to graphically interpret signals in both time and frequency domains.
3. Students should become familiar with common signal types such as **sinusoidal signals, exponential signals, unit step, and impulse**, since these will appear repeatedly throughout the course.

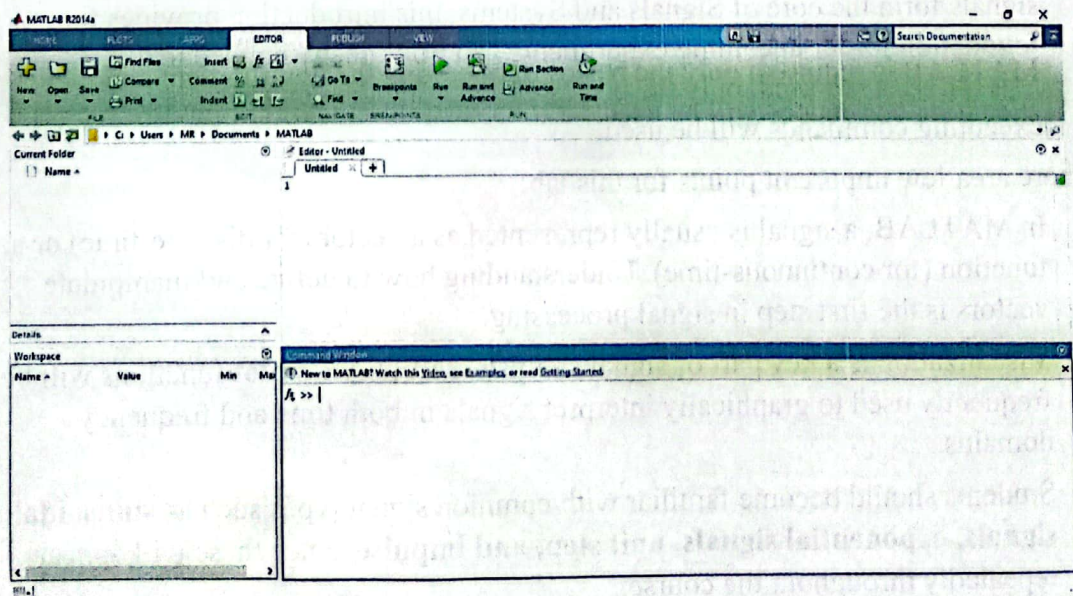


Working Environment (2014a)



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Working Environment (2014a)



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Variables

Example
Compute the expressions

- $\sin(45^\circ)^2 + \cos(45^\circ)^2$
- $\sin(45^\circ) \cdot \cos(45^\circ)$

Commands	Results	Comments
<code>theta=45*pi/180</code>	<code>theta = 0.7854</code>	First, we convert the 45° to radians. The result is assigned to a variable named <i>theta</i> .
<code>a=sin(theta)</code>	<code>a = 0.7071</code>	Definition of the variable <i>a</i> .
<code>b=cos(theta)</code>	<code>b = 0.7071</code>	Definition of the variable <i>b</i> .
<code>c=a^2+b^2</code>	<code>c = 1</code>	The result of the first expression is assigned to the variable <i>c</i> .
<code>a*b</code>	<code>ans = 0.5000</code>	The second expression is computed and the result is stored in the default variable <i>ans</i> .
<code>d=c+ans</code>	<code>d = 1.5000</code>	The variable <i>ans</i> can be used for further computations.
<code>A=6</code>	<code>A = 6</code>	Definition of the variable <i>A</i> .
<code>a=3</code>	<code>a = 3</code>	Definition of the variable <i>a</i> .
<code>c=A+a</code>	<code>c = 9</code>	It is obvious that there is distinction between small and capital letters.



Variables

Commands	Results	Comments
<code>d=A+a;</code>		Nothing is displayed at the command window. This of course does not mean that the command is not executed.
<code>d</code>	<code>d = 9</code>	To see the result simply type the variable to which it is assigned.

Commands	Results	Comments
<code>x='hello'</code>	<code>x = hello</code>	Text assignment to a variable.

The text is inserted between **single quotes**.



Vectors (Row Vectors)

Commands	Results	Comments
<code>a = [1 2 3 4 5]</code>	<code>a = 1 2 3 4 5</code>	Row vector <i>a</i> of 5 elements.
<code>b = [-6, -7, -8, -9, -10]</code>	<code>b = -6 -7 -8 -9 -10</code>	Row vector <i>b</i> of 5 negative elements.
<code>c = [0.3, 8.5, 6, -2.4]</code>	<code>c = 0.30 8.50 6.00 -2.40</code>	Row vector <i>c</i> of 4 decimal elements.
<code>b(3)</code>	<code>ans = -8</code>	The third element of vector <i>b</i> .
<code>b(6)</code>	<code>??? Index exceeds matrix dimensions.</code>	Vector <i>b</i> has 5 elements. Thus, referring to its 6th position causes an error.



1.6.2 Commands length/size

Commands	Results	Comments
<code>length(a)</code>	<code>ans = 5</code>	Number of elements of vector <i>a</i> .
<code>size(a)</code>	<code>ans = 1 5</code>	Vector <i>a</i> consists of 1 row and 5 columns.

Commands	Results	Comments
<code>a</code>	<code>a = 1 2 3 4 5</code>	Vector <i>a</i> is already defined.
<code>a(3) = 99</code>	<code>a = 1 2 99 4 5</code>	The value 99 is inserted at the third position of vector <i>a</i> , replacing the previous value.
<code>a(10) = 23</code>	<code>a = 1 2 99 4 5 0 0 0 0 23</code>	The value 23 is inserted at the tenth position of <i>a</i> . Notice that the intermediate (6th, 7th, 8th, and 9th) positions are filled with zeros.



Commands length/size

Commands	Results	Comments
<code>a=1:2:11</code> start, step, stop	<code>a = 1 3 5 7 9 11</code>	The elements of vector <i>a</i> start from 1, end at 11, and are spaced by 2.
<code>b=1:6</code>	<code>b = 1 2 3 4 5 6</code>	If no step is specified, by default the step considered is 1.
<code>c=0:0.2:1</code>	<code>c = 0 0.2 0.4 0.6 0.8 1.0</code>	Decimal step.
<code>d=3:-1:-3</code>	<code>d = 3 2 1 0 -1 -2 -3</code>	Negative step.
<code>e=0:pi/6:pi</code>	<code>e = 0 0.5236 1.0472 1.5708 2.0944 2.6180 3.1416</code>	Vector <i>e</i> is defined from 0 to π with step $\pi/6$.

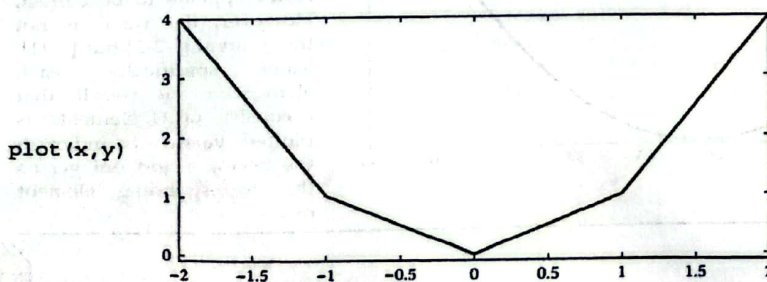
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Plotting in Two Dimensions

Example

Plot the function $y(x) = x^2$, $-2 \leq x \leq 2$. `[-2, 4], [-1, 1], [0, 0], [1, 1], [2, 4]`

Commands	Results	Comments
<code>x=-2:2</code>	<code>x = -2 -1 0 1 2</code>	Vector <i>x</i> is defined from -2 to 2 with step 1.
<code>length(x)</code>	<code>ans = 5.00</code>	Vector <i>x</i> has 5 elements.
<code>y=x.^2</code>	<code>y = 4 1 0 1 4</code>	Vector <i>y</i> is created. Notice that $y = [(-2)^2 (-1)^2 0^2 1^2 2^2]$. Also notice the dot operator which is inserted before the power operator to ensure that this is an element per element operation.
<code>length(y)</code>	<code>ans = 5.00</code>	Vector <i>y</i> has the same number of elements to <i>x</i> .



Graph of the function $y(x) = x^2$, $-2 \leq x \leq 2$.

Plotting in Two Dimensions

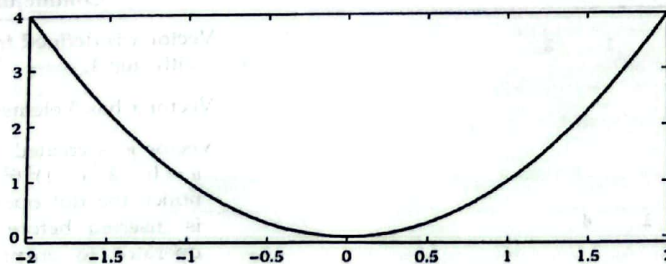
Commands	Results	Comments
$x = -2:0.1:2$	$x = -2.00 \quad -1.90 \quad -1.80 \quad -1.70 \quad -1.60$ $-1.50 \quad -1.40 \quad -1.30 \quad -1.20 \quad -1.10$ $-1.00 \quad -0.90 \quad -0.80 \quad -0.70 \quad -0.60$ $-0.50 \quad -0.40 \quad -0.30 \quad -0.20 \quad -0.10$ $0 \quad 0.10 \quad 0.20 \quad 0.30 \quad 0.40 \quad 0.50 \quad 0.60$ $0.70 \quad 0.80 \quad 0.90 \quad 1.00 \quad 1.10 \quad 1.20$ $1.30 \quad 1.40 \quad 1.50 \quad 1.60 \quad 1.70 \quad 1.80$ $1.90 \quad 2.00$	The vector x is defined from -2 to 2 with step 0.1.
<code>plot(x,y)</code>	??? Error using $\Rightarrow$ plot Vectors must be the same lengths.	Trying to execute the command plot we get an error warning as $\text{length}(x) = 41$ and $\text{length}(y) = 5$.

Plotting in Two Dimensions

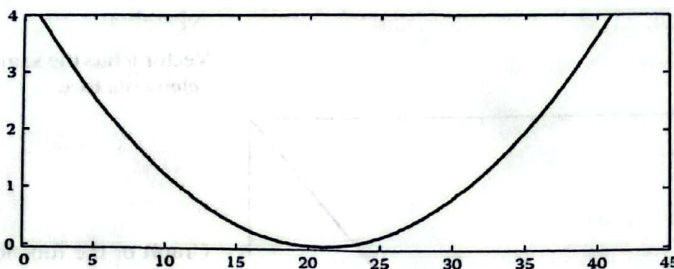
`y = x.^2;`

`length(y)` ans = 41.00

`plot(x,y)`



`plot(y)`



Vector y has to be redefined according to the new vector x .

Now the two vectors have the same number of elements and the function $y(x) = x^2$, $-2 \leq x \leq 2$ can be plotted.

This time the graph is satisfactory.

It is possible to execute the command `plot` with only one input argument. The result appears to be correct. However, the x -axis is not the interval $[-2 \ 2]$ but $[1 \ 41]$. More specifically, each element of y (recall that y consists of 41 elements) is plotted versus its index in the vector y and not versus the corresponding element of x .

The Fig File

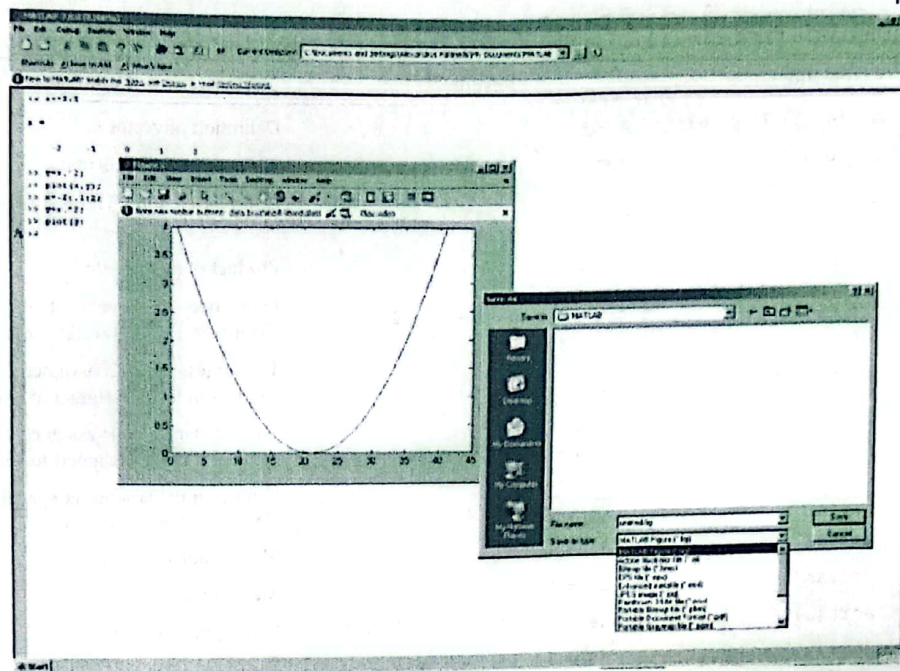
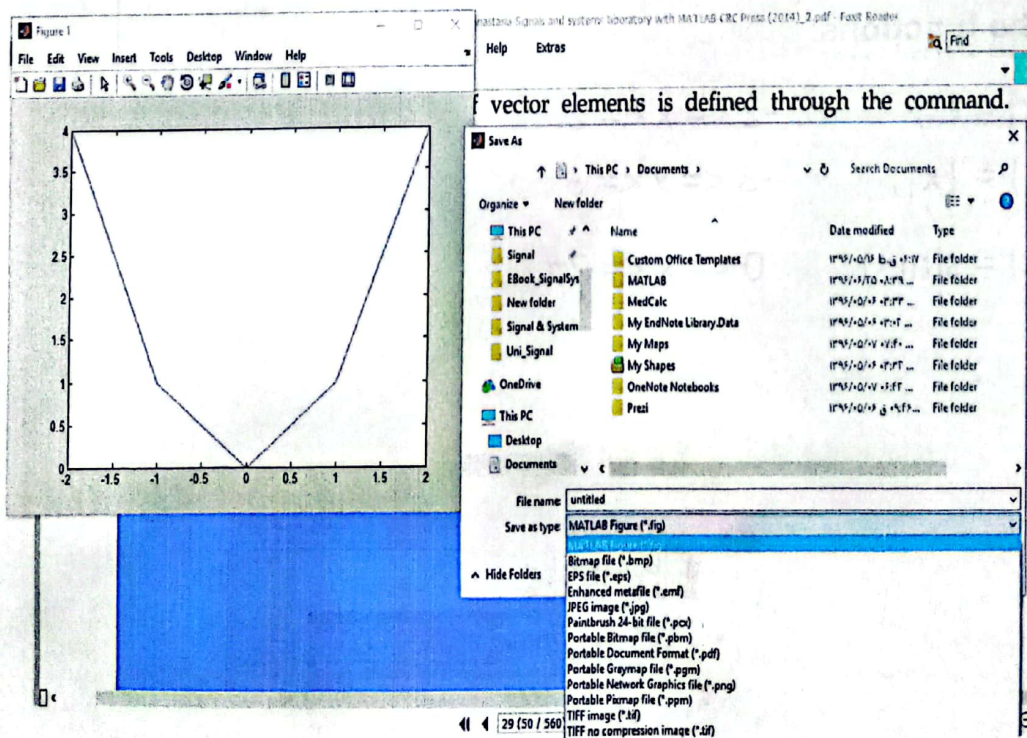


FIGURE 1.2
A figure can be exported from MATLAB in the most common formats.

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1.8.2 The Fig File



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1.6.7 Useful Commands

sum-cumsum-prod-diff-max-min-sort-mean-median

Commands	Results	Comments
<code>a = [4 2 7 0 6]</code>	<code>a = 4 2 7 0 6</code>	Definition of vector <i>a</i> .
<code>s = sum(a)</code>	<code>s = 19</code>	Sum of the elements of <i>a</i> .
<code>c = cumsum(a)</code>	<code>c = 4 6 13 13 19</code>	Cumulative sum. The result is obtained as [4, 4+2, 4+2+7, 4+2+7+0, 4+2+7+0+6]. تراكمي
<code>p = prod(a)</code>	<code>p = 0</code>	Product of all elements. متتالي مرتبط
<code>d = diff(a)</code>	<code>d = -2 5 -7 6</code>	Difference between two consecutive elements, i.e., $d(1) = a(2) - a(1)$, etc.
<code>[m, i] = max(a)</code> index	<code>m = 7</code> <code>i = 3</code>	The largest value is assigned to variable <i>m</i> , and its index is assigned to variable <i>i</i> .
<code>[m, i] = min(a)</code>	<code>m = 0</code> <code>i = 4</code>	The smaller value is assigned to variable <i>m</i> , and its index is assigned to variable <i>i</i> .
<code>max(a)</code>	<code>ans = 7</code>	If no output variable is specified, only the largest value is returned.
<code>mean(a)</code>	<code>ans = 3.8000</code>	Mean value of the elements.
<code>median(a)</code>	<code>ans = 4</code>	Median value of the vector.
<code>sort(a)</code>	<code>ans = 0 2 4 6 7</code>	Sorting in ascending order.
<code>sort(a, 'descend')</code>	<code>ans = 7 6 4 2 0</code>	Sorting in descending order.

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Assignments

Plot the functions:

1- $Y(x) = e^x$, $-2 \leq x \leq 2$

2- $y(x) = |x|$, $-3 \leq x \leq 3$

3- $Y(x) = \sin(x)$, $0 \leq x \leq 2\pi$



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